SENTHIL PUBLIC SCHOOL, DHARMAPURI

AFFILIATED TO CBSE, NEW DELHI (AFFL. NO. 1930248)

CLASS: CLASS 8 (CBSE)	**SUBJECT: MATHEMATICS**	**DATE: 18-OCT-2026**
TIME: 60 MINS	**PHASE: Phase 1: Comprehensive Weekly Sunday Examination**	**MAX. MARKS: 35**

--- SECTION A — OBJECTIVE & ASSERTION-REASONING ---

Q.1 Which of the following algebraic identities is universally correct for all real values of a and b? **[1]**
(a) $(a + b)^2 = a^2 + b^2$ (b) $(a - b)^2 = a^2 - 2ab + b^2$ (c) $(a + b)(a - b) = a^2 + b^2$ (d) $(a + b)^2 = a^2 - 2ab + b^2$

Q.2 Pressure is defined as the force acting per unit area of a surface. Its standard S.I. unit is: **[1]**
(a) Newton (N) (b) Joule (J) (c) Pascal (Pa) (d) Watt (W)

Q.3 Which agricultural method is recognized as the most water-efficient modern irrigation system? **[1]**
(a) Moat (Pulley system) (b) Rahat (Lever system) (c) Drip Irrigation (d) Chain Pump

Q.4 The British land revenue settlement that fixed the revenue demand permanently without periodic revision was: **[1]**
(a) Ryotwari Settlement (b) Mahalwari Settlement (c) Permanent Settlement (1793) (d) Zabt System

Q.5 ASSERTION (A): The shoulder strap of a school bag is made broad. REASON (R): A broad strap increases the surface area of contact, thereby reducing the pressure exerted on the shoulder. **[1]**
(a) Both (A) and (R) are true and (R) is the correct explanation of (A). (b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A). (c) (A) is true but (R) is false. (d) (A) is false but (R) is true.

--- SECTION B — VERY SHORT ANSWER QUESTIONS ---

Q.6 Factorise the algebraic expression completely by grouping terms: $15xy - 6x + 5y - 2$. **[2]**

Q.7 State two distinct physiological differences between Plant Cells and Animal Cells regarding cell wall and vacuole size. **[2]**

Q.8 Why is rolling friction substantially less than sliding friction? Give one practical engineering application where sliding friction is replaced by rolling friction. **[2]**

--- SECTION C — SHORT ANSWER & NUMERICAL QUESTIONS ---

Q.9 A wooden block of dimensions 20 cm x 10 cm x 5 cm of mass 4 kg is kept on a tabletop. Calculate the maximum pressure exerted by the block on the tabletop in Pascals. (Take g = 10 m/s²). **[3]**

Q.10 The perimeter of a rectangular swimming pool is 154 m. Its length is 2 m more than twice its breadth. Find the length and breadth of the pool step-by-step. **[3]**

Q.11 Explain three major economic causes that triggered widespread discontent among Indian peasants and artisans during the Revolt of 1857. **[3]**

--- SECTION D — LONG ANSWER & CASE-BASED INTEGRATED TASKS ---

Q.12 (a) A closed cylindrical storage tank of radius 7 m and height 3 m is made from a sheet of metal. How much sheet of metal is required? (Take $\pi = 22/7$). (b) If the metal costs ☐ 45 per square meter, calculate the total cost of metal used. **[5]**

Q.13 (a) Draw a neat labeled diagram of a Plant Cell showing at least 4 key organelles. (b) State the specific function of: (i) Mitochondria, and (ii) Chloroplast. **[5]**

Q.14 CASE STUDY: An organic farmer in Dharmapuri tests two identical soil plots. Plot A is treated with biological compost and cow dung manure. Plot B is treated with continuous chemical urea and ammonium sulfate. (i) State two long-term ecological advantages observed in Plot A regarding soil texture and water retention. [2] (ii) Why does excessive chemical fertilizer usage degrade soil fertility over multiple seasons? [2] (iii) What biological process restores nitrogen naturally to agricultural soil? [1] **[5]**